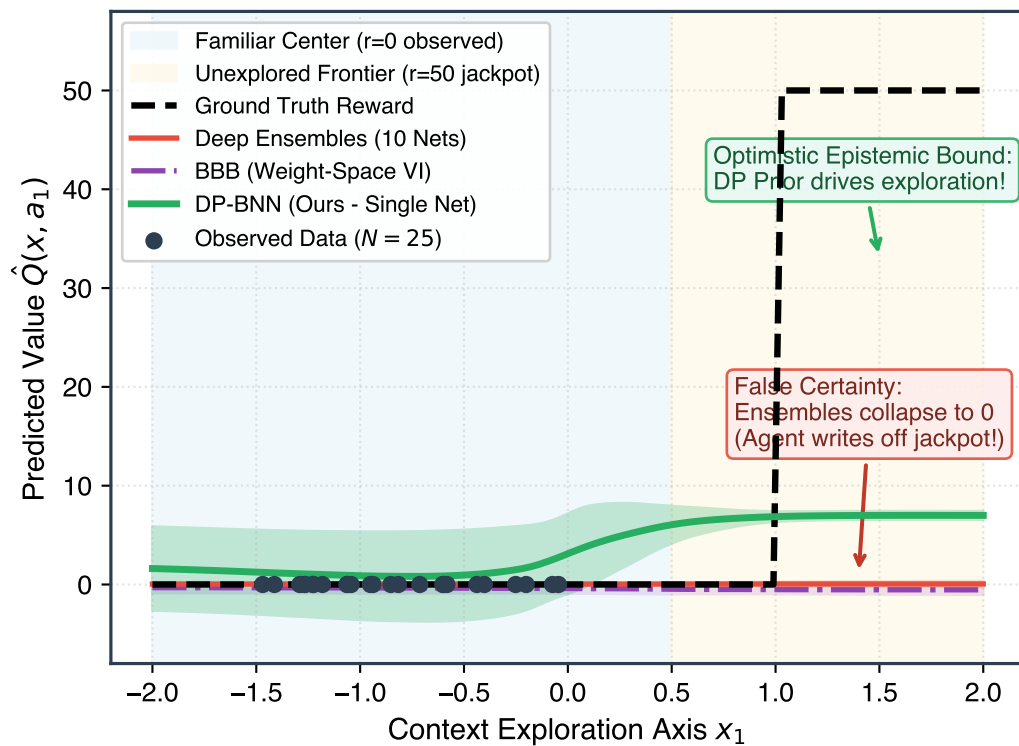


Why Measure-Space Dirichlet Priors Conquer Deceptive Sequential Decision-Making

[The DP-BNN Single-Network Advantage vs. Weight-Space Prior Collapse]

A. Value Prediction on Unexplored Frontier $\hat{Q}(x, a_1)$



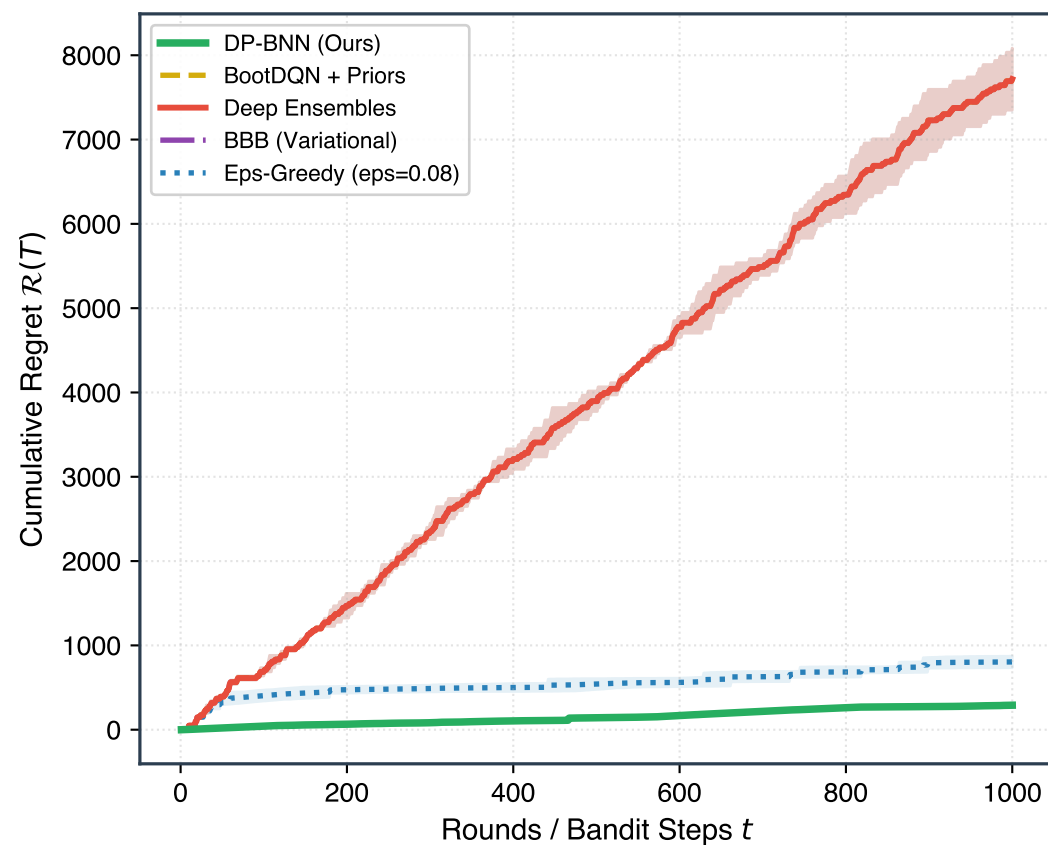
Weight-Space Paradigm (Ensembles / BBB)

- Priors on Weights: $w \sim \mathcal{N}(0, \sigma^2 I)$
- Memory Overhead: 8–20 separate neural networks
- Inference: Independent SGD runs or Gaussian VI
- Fatal Flaw in Decision-Making:
 - Activations align strictly to observed center data ($r = 0$).
 - Weight-space noise cancels out across deep ReLUs.
 - > Predicts false certainty (Q) in void!
 - > Permanently trapped in deceptive safe arm.

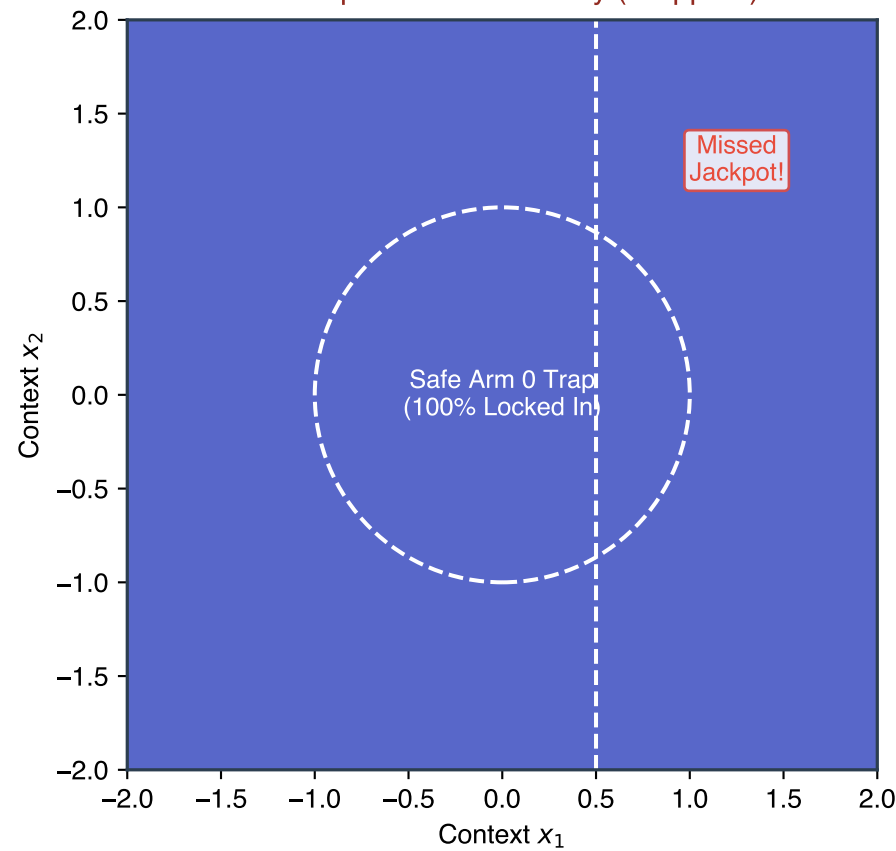
Measure-Space Paradigm: DP-BNN (Ours)

- Prior on Data Measure: $\mathcal{P} \sim \text{DP}(\alpha, P_0)$
- Single Network Efficiency: 1 single NN in memory
- Conjugate Posterior: $\mathcal{P} \mid \mathcal{D} \sim \text{DP}(\alpha + N, \frac{\alpha P_0 + \sum \delta_{z_i}}{\alpha + N})$
- Principled Exploration Mechanism:
 1. Center observations receive empirical mass $N/(N + \alpha)$.
 2. Unexplored frontier retains base measure prior mass $\alpha/(N + \alpha)$.
 3. Dirichlet weights automatically generate optimistic hypotheses!--> Agent explores frontier and discovers $r = 50$ jackpot!

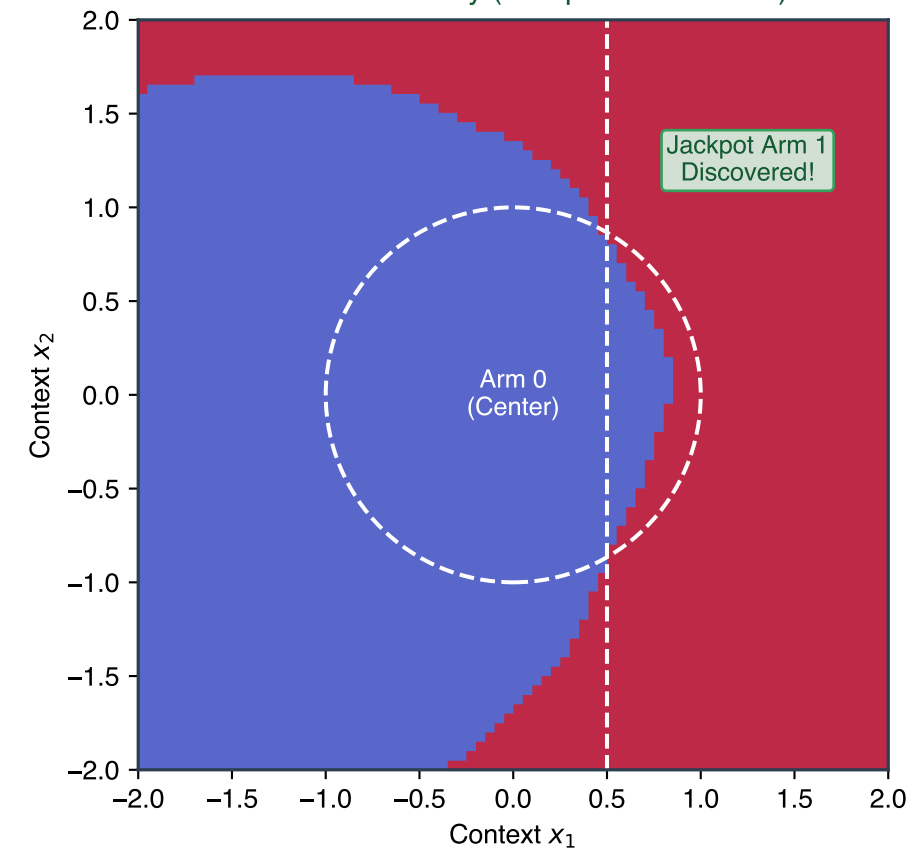
C. Cumulative Regret over 1000 Rounds (Lower is Better)



D. Deep Ensembles Policy (Trapped!)



E. DP-BNN Policy (Jackpot Discovered!)



Arm 0: Deceptive Safe Trap ($r = 1$)

Arm 1: Hidden Jackpot Frontier ($r = 50$)